

Harish Manoharan

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Software engineer building production LLM systems (retrieval, model evaluation, AI agents) at Purgo AI since 2025.
M.S. Computer Science, UT Dallas (GPA 3.92/4.00); dual degree in Data Science and Biological Sciences, IIT Madras.

EXPERIENCE

Purgo AI

San Francisco Bay Area (Los Altos, CA)

AI startup building an agent that writes and validates Databricks and dbt code

Software Engineer

Jun 2026 – Present

Software Engineer Intern

Jun 2025 – May 2026

- Cut the LangGraph agent's missed source tables by 70% (79 to 24 on 103 tickets) by building dbt lineage expansion with sqlglot; chose the design in a follow-up four-variant ablation and rejected repo search (4.7x more unwanted tables).
- Ran a 5-run, 102-ticket comparison showing a smaller reasoning model lost no retrieval recall (0.944 vs 0.931 for the larger model), then moved the agent's retrieval and drafting stages to it, cutting their LLM cost 7.6x.
- Cut mean catalog-search latency from 4.0 s to 0.18 s and raised recall@10 from 0.54 to 0.68 vs the existing search path by building Pinecone semantic search for Polaris/Iceberg catalogs (incremental indexing) and its 188-query benchmark.
- Built an LLM evaluation and validation product end to end that qualifies Azure OpenAI and Databricks-served models for GxP use: 19 tests for hallucination on missing-field inputs, accuracy/F1, bias, determinism, p95 latency, traceability.
- Built most of the IQ/OQ (installation and operational qualification) engine for Databricks on Azure and AWS: 59 versioned tests, sandboxed pass/fail logic, auditor evidence reports and an LLM pipeline turning requirement docs into tests.
- Extended the self-review loop in the agent's LangGraph orchestration: AST lint rules for generated PySpark/SQL, each traced to a real Databricks runtime failure, plus recovery regeneration and best-candidate selection.

Digitus Business Solutions

Bengaluru, India

Software Engineer Intern

May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building a text-to-SQL agent with schema discovery and retries driven by execution errors.
- Built a multimodal RAG assistant over internal slide decks and videos (layout-aware chunking, enforced citations).

PwC India

Bengaluru, India

Technology Consulting Intern, Data Analytics

May 2023 – Jul 2023

- Cut forecast error (MAPE) 15% against the baseline across 50+ SKUs with an XGBoost + SARIMA ensemble; built an aspect-based sentiment pipeline over 30k+ customer reviews.

EDUCATION

The University of Texas at Dallas M.S. Computer Science

May 2026

GPA 3.92/4.00 · Dean's Graduate Scholar

Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences

May 2024

INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Machine learning and LLMs: retrieval-augmented generation (RAG), LLM evaluation, AI agents (LangGraph), embeddings and vector search (Pinecone, pgvector), LLM observability (Langfuse), PyTorch

Data: Databricks, Apache Spark (PySpark), dbt, PostgreSQL, Apache Iceberg

Backend and cloud: FastAPI, NestJS, Next.js, REST APIs, GCP, AWS, Docker, CI/CD (GitHub Actions)